



WGS

WINDOWS GAME SERVER

Single-window management panel for Windows game servers

.NET 8 · WPF · SteamCMD · 17+ games

17+ Games

Ready-made plugins

SteamCMD

Install & update

Auto Restart

Crash recovery

Dashboard

Live metrics

Firewall

Auto rules

Backups

Zip + restore

RCON

Remote console

CPU Affinity

Core selection

Encrypted

DPAPI security

Free, open-source Windows game server manager — no command-line required.
Supports 17+ games with SteamCMD integration, backups, firewall, RCON and more.

1. Introduction

Windows Game Server (WGS) is a free, open-source management application that makes installing, running and managing Windows game servers straightforward. Every essential operation is available from a single window — no command-line knowledge required for day-to-day use.

Technology	.NET 8 · WPF · CommunityToolkit.Mvvm · SteamCMD · Microsoft.Extensions.DependencyInjection
Requirements	Windows 10 / Server 2019 or newer · .NET 8 Runtime · Administrator rights for firewall management

2. Installation

2.1 Pre-built binary (recommended)

1	Download Grab the latest release zip from the GitHub Releases page.
2	Extract Extract the zip to any folder, e.g. C:\WGS\
3	Run Double-click WindowsGameServer.exe — the app starts immediately.
4	.NET Runtime If you see a runtime error, install .NET 8 Runtime from dotnet.microsoft.com
5	SteamCMD WGS downloads SteamCMD automatically the first time you install a game.

2.2 Build from source

Requires .NET 8 SDK.

```
git clone https://github.com/YOUR_USERNAME/WindowsGameServer.git
cd WindowsGameServer/WGS
dotnet publish -c Release -o publish
```

3. First Launch

When the app opens for the first time you will see an empty server list. The left sidebar contains:

- **Dashboard** — global system metrics and server status summary
- **Server list** — all configured servers with their current status
- **+ Add Server** — opens the new server creation dialog
- **Icon toolbar** — quick access to Settings, Discord, Web API, Workshop, Mods and more

Tip

The system-tray icon (bottom-right corner): WGS stays running when you close the window. Double-click the tray icon to reopen.

4. Adding a Server

- 1 Open dialog**
Click the '+ Add Server' button in the left sidebar.
- 2 Select game**
Pick from 17+ pre-configured games in the dropdown.
- 3 Name**
Give the server a recognisable name (e.g. 'Valheim — Family').
- 4 Install folder**
WGS suggests a path automatically. Change it if needed.
- 5 Create**
Click 'Create server' — it appears with status 'Not Installed'.

Note

No files are downloaded here — only the server definition is created. Use 'Install / Update' to download game files.

5. Install & Update

WGS uses Valve's SteamCMD to download game files. SteamCMD is fetched automatically into WGS's own AppData folder on first use.

- 1 Select server**
Click a server in the left list.
- 2 Install / Update**
Click 'Install / Update' in the server detail view.
- 3 Login**
Games requiring a purchased copy will ask for Steam credentials. Most games support anonymous login.

4 Progress
The console shows real-time download percentage from SteamCMD.

5 Done
When the download finishes the status changes to 'Stopped'.

Steam Guard

If Steam Guard is enabled, SteamCMD will request an email code on first login. Enter the code in the console input field.

6. Starting & Stopping

Control buttons in the server detail view:

Button	Action	Notes
Start	Launches the server process	Runs a port pre-flight check first
Stop	Sends graceful stop command; kills after 5 s	Waits for clean shutdown
Kill	Kills the entire process tree immediately	No grace period
Restart	Stop → 3 s delay → Start	
Show Window	Brings the server console window to the front	Only if the process has a window

Status colours

Status	Colour	Meaning
Running	Green	Server is running normally
Starting	Blue	Process is starting up
Stopping	Orange	Graceful shutdown in progress
Stopped	Grey	Stopped — game files are installed
Not Installed	Grey	Game files have not been downloaded yet
Error	Red	Process failed to start or crash limit reached

7. Automation & Auto-Start

- **Auto Start** — starts the server automatically when WGS is launched
- **Auto Restart** — restarts the server after a crash with a configurable delay (default 10 s)
- **Auto Restart Max Retries** — if the server crashes more than N times within 10 minutes, auto-restart is suspended to prevent an infinite crash loop (default 5)
- **Auto Update** — runs Install/Update on a configurable interval (minutes) while the server is running; stops and restarts the server automatically

Crash loop detection

WGS counts crashes within a rolling 10-minute window. If the count exceeds Max Retries, auto-restart is disabled and an error is logged. Re-enable Auto Restart manually after fixing the underlying problem.

Auto Update interval

Set 'Auto Update Interval (min)' in the server Settings tab. 0 disables periodic updates. A backup is created automatically before each update if Backup Enabled is checked.

8. Server Settings

8.1 General fields

Field	Description
Display Name	Internal name used inside WGS
Server Name	Public name shown in the in-game server browser
Max Players	Maximum allowed player count
Server IP	Listen address (0.0.0.0 = all network interfaces)
Game Port	Main port — players connect here
Query Port	Steam Query port used by the server browser
Steam Port	Steam network port (required by some games, e.g. Wreckfest)
RCON Port	Remote console port (Source RCON protocol)
RCON Password	RCON password — do not leave blank on a public server
Server Password	Connection password for players (optional)
GSLT Token	Steam Game Server Login Token (requires a Steam account)
Custom Args	Extra command-line arguments appended to the start command
Install Path	Folder where the game files live on disk

8.2 Performance settings

Each server can be restricted to specific CPU cores and assigned a Windows process priority. Changes apply on the next startup.

Setting	Values	Description
CPU Affinity	Core checkboxes	Which CPU cores the server may use. All unchecked = no restriction
Process Priority	Normal / AboveNormal / High / BelowNormal / RealTime	Windows process priority. High and RealTime require admin rights.

Warning

RealTime priority can freeze the entire system if the server runs at 100 % CPU. Use with caution.

9. Firewall

WGS manages Windows Firewall rules automatically when 'Firewall Auto-Manage' is enabled (default: on).

- On server **start**: WGS adds inbound TCP + UDP rules for game, query and Steam ports.
- On server **stop**: those rules are removed automatically.
- If Auto-Manage is **off**: add rules manually in Windows Defender Firewall.
- Rules use **netsh advfirewall** — administrator rights are required.

10. Ports & Networking

Before startup WGS checks whether the required ports are free locally. A port-in-use warning is logged but startup continues.

Port type	Purpose	Default
Game Port	Players connect to this port	Varies by game
Query Port	Steam server browser queries this port	Usually Game Port + 1
Steam Port	Steam network traffic (e.g. Wreckfest)	27015
RCON Port	Remote console (Source RCON protocol)	Varies

Router / NAT

To make the server reachable from the internet you must set up Port Forwarding on your router. WGS's pre-flight check only detects local conflicts.

11. Console & RCON

11.1 Console

The Console tab shows server output in real time. Messages are colour-coded:

Colour	Type	Description
White / Grey	Info	Normal log messages
Yellow	Warning	Lines containing 'warn'
Red	Error	Lines containing 'error', 'exception' or 'fatal'
Cyan	System	WGS messages and SteamCMD output

Filter

The Filter field above the console searches the log in real time (case-insensitive).

11.2 RCON remote console

RCON is the Valve Source protocol for sending admin commands over the network. Works with Rust, 7 Days to Die, and other Source-based games.

1

Set RCON port & password

Fill in 'RCON Port' and 'RCON Password' in the server's Settings tab.

2

Start the server

RCON becomes available when the server process is running.

3

Connect

Click 'RCON: Connect' in the Console tab.

4

Send a command

Type the command and press Enter or click 'Send'.

12. Config Editor

The **Config** tab lets you browse and edit server configuration files directly inside WGS — no file manager or text editor needed.

- Click the **Config** tab in the server detail view.
- WGS scans the install folder and lists all recognised configuration files.
- Click a file name to load its contents into the built-in text editor.
- Edit the content, then click **Save** to write changes to disk.
- Click **Open folder** (📁) to open the install directory in Windows Explorer.

Tip

Stop the server before editing files it holds open. Changes take effect on the next start.

13. Performance Charts

The **Charts** tab displays real-time CPU and RAM usage history for the selected server while it is running.

- **CPU % chart** — process CPU usage as a percentage of total system CPU capacity
- **RAM chart** — working-set memory consumption in MB
- History window: last **6 minutes** — 180 data points sampled every 2 seconds
- Charts reset automatically when the server is stopped

Note

CPU % is normalised across all logical cores. A server fully using one core on a 4-core machine shows ~25 %.

14. Mod Manager

The **Mods** tab manages server-side modding frameworks. Two frameworks are supported:

- **Oxide / uMod** — for Rust and other supported games. Downloads the latest Oxide release and installs it into the server folder.
- **Minecraft plugins** — for Paper/Spigot servers. Displays the contents of the plugins folder.

Action	Description
Install Oxide	Downloads and installs the latest Oxide/uMod release for this game
Update Oxide	Re-downloads Oxide to update to the latest version
Open plugins folder	Opens the server's plugins directory in Windows Explorer

Note

Stop the server before installing or updating Oxide. Oxide injects into the server process on startup.

15. Steam Workshop

The **Workshop** tab is visible for games that support Steam Workshop (e.g. Arma Reforger). It uses SteamCMD to download Workshop items into WGS's steamapps workshop folder.

- Find the item ID**
Open the Workshop page in a browser. The ID is the number in the URL:
`steamcommunity.com/sharedfiles/filedetails/?id=XXXXXXXXX`
- Enter the ID**
Paste the Workshop item ID into the input field in the Workshop tab.
- Download**
Click 'Download'. WGS runs SteamCMD in the background and shows progress in the console.
- Done**
The item appears in the Installed Items list with its name and local folder path.

- Click **Remove** next to an item to delete it from disk.
- Workshop content is stored in: `%APPDATA%\WGS\steamcmd\steamapps\workshop\content\`

16. Player Statistics

The **Players** tab shows session history for everyone who has connected to the server. Data is stored in a local SQLite database and survives restarts.

Column	Description
Player Name	Name as reported in the server log
Join Time	Timestamp when the player connected
Leave Time	Timestamp when the player disconnected (blank if still online)
Duration	Total time in session — format HH:MM:SS

- The **Total Playtime** view shows cumulative playtime per player across all sessions.
- Join/leave detection works by parsing the server's log output — accuracy depends on each game's log format.
- Stored in: `%APPDATA%\WGS\player_stats.db`

17. Backups

WGS creates zip archives of a server's save data. Backups are configured per server.

- **Backup Enabled** — activates automatic backups for this server
- **Backup Retention** — number of backups to keep (default 5); older ones are deleted automatically
- **Create Backup** button — takes an immediate backup
- **Backup All** (Dashboard) — backs up every server at once
- **Restore** — extracts a backup into the install folder (stop the server first)
- Stored in: `%APPDATA%\WGS\backups\<server-id>\`

Disk space	Large servers (ARK, Valheim) produce big backups. Set a sensible retention value and monitor free disk space.
Auto backup before update	If Backup Enabled is checked, WGS creates a backup automatically before every Install/Update and before every Auto Update cycle.

18. Scheduled Tasks

The built-in scheduler runs actions automatically at set times. Add tasks in the **Schedule** tab of any server.

Field	Options / Description
Action	Restart · Stop · Start · Backup · Update
Frequency	Once · Daily · Weekly
Time of day	HH:MM (24-hour clock)
Day of week	Monday–Sunday (Weekly only)
Enabled	Toggle a task on/off without deleting it
Last / Next	Timestamps — Next run is calculated automatically

Polling interval

The scheduler checks every 30 seconds, so actual execution may be up to 30 s after the configured time.

19. Dashboard

Click **Dashboard** in the sidebar to open the live system overview. Data refreshes every 2 seconds.

- **CPU Usage** — overall processor load as a percentage
- **RAM** — used and free memory in GB with percentage bar
- **Drives** — usage and free space for every connected drive
- **Server Count / Online / Stopped** — server status summary
- **Backup All** — triggers an immediate backup of every configured server

20. Discord Notifications

WGS posts rich embed messages to a Discord channel via a webhook URL. Configure in **Settings** → **Discord Notifications**.

Event	Trigger	Embed colour
Server started	Status → Running	Green
Server stopped	Status → Stopped	Red
Server crashed	Status → Error	Red
Update started	Status → Updating	Blue

- Create webhook**
Discord: channel Settings → Integrations → Webhooks → New Webhook → Copy URL.
- Paste URL**
Open WGS Settings → Discord Notifications → paste into 'Webhook URL'.
- Enable**
Check 'Enable Discord notifications' and choose which events to send.
- Test**
Click 'Test' to send a test embed immediately.
- Save**
Click 'Save'. The URL is encrypted with Windows DPAPI before writing.

21. Discord Remote Control Bot

The Discord bot lets you control your servers interactively from any Discord channel. Unlike the notification webhook, the bot listens for commands and replies with results. Configure in **Settings** → **Discord Remote Control Bot**.

- Create a bot**
Go to discord.com/developers/applications → New Application → Bot → Reset Token → Copy Token.
- Set permissions**
Under OAuth2 → URL Generator select 'bot' scope and 'Read Messages / View Channels' + 'Send Messages' permissions. Invite the bot to your server.
- Get channel ID**
Enable Developer Mode in Discord (Settings → Advanced). Right-click your control channel → Copy Channel ID.

4	Configure WGS Open WGS Settings → Discord Remote Control Bot. Paste the Bot Token and Channel ID. Optionally set a custom command prefix (default: !).
5	Restrict access Enter comma-separated Discord User IDs in 'Allowed User IDs' to limit who can send commands. Leave blank to allow everyone in the channel.
6	Save & test Click Save, then 'Test connection'. The bot posts a message in the channel if working.

Bot commands

Command	Description
!help	Lists all available commands
!status	Shows all servers with their current status
!start <name>	Starts the named server
!stop <name>	Stops the named server
!restart <name>	Restarts the named server
!update <name>	Runs Install/Update on the named server
!backup <name>	Creates an immediate backup of the named server
!cmd <name> <command>	Sends a console command to the running server

Command prefix	The default prefix is '!'. Change it in Settings → Command prefix. Server names are matched case-insensitively.
Security	Keep the bot token secret — treat it like a password. Use Allowed User IDs on public servers to prevent unauthorised control.

22. Web API

WGS includes a built-in HTTP server that exposes a REST API and a browser-based management UI. Enable it in **Settings** → **Web Remote Control**.

1	Enable Check 'Enable Web API' in Settings and choose a port (default: 8765).
2	Token An access token is generated automatically. Copy it or enter your own.
3	Save Click Save — the API starts immediately and the status indicator turns green.

4

Open browser UI
Navigate to `http://localhost:8765/ui` — a management page loads with server controls.

5

Remote access
Use your machine's LAN IP (e.g. `http://192.168.1.100:8765`) from other devices. For internet access, add a port forwarding rule on your router.

REST endpoints

Method	Endpoint	Description
GET	<code>/api/servers</code>	List all servers with status and port info
GET	<code>/api/servers/{id}</code>	Get a single server's details
POST	<code>/api/servers/{id}/start</code>	Start a server
POST	<code>/api/servers/{id}/stop</code>	Stop a server
POST	<code>/api/servers/{id}/restart</code>	Restart a server
POST	<code>/api/servers/{id}/update</code>	Run Install/Update
POST	<code>/api/servers/{id}/backup</code>	Create a backup
POST	<code>/api/servers/{id}/cmd</code>	Send console command (body: <code>{"command":"..."}</code>)
GET	<code>/api/system</code>	System metrics — CPU, RAM, drives

Authentication	All <code>/api/*</code> endpoints require the token via the <code>Authorization: Bearer</code> header or the <code>?token=</code> query parameter.
CORS	All origins are allowed — the API can be called from any web page or external tool.

23. Supported Games

Survival

Game	AppID	Max Players	Default Port
Valheim	896660	10	2456
Rust	258550	100	28015
7 Days to Die	294420	8	26900
Conan Exiles	443030	40	7777
ARK: Survival Evolved	376030	70	7777
Sons of the Forest	2465200	8	8766
The Forest	556450	8	27015
Survive the Nights	1502300	16	7777
SCUM	3792580	32	10000
Vein	2131400	16	7777

Racing

Game	AppID	Max Players	Default Port
Wreckfest	361580	24	33540
Wreckfest 2	3519390	24	27020
Assetto Corsa	302550	18	9600

Other

Game	AppID	Max Players	Default Port
Minecraft Java	—	20	25565
Euro Truck Simulator 2	1948160	8	27015
Arma Reforger	1874900	64	2001
Black Mesa	346680	24	27015

24. Custom Plugin Creator

The Plugin Creator lets you add any game server without writing code. Open it from **Tools** → **Plugin Creator**.

Field	Req.	Description
Game ID	Yes	Unique identifier, e.g. 'mygame'
Game Name	Yes	Display name in the game dropdown
Description	No	Short description shown in the Info tab
Category	No	Survival / Racing / Shooter / etc.
Steam AppID	Yes	Used by SteamCMD to download the server files
Executable	Yes	Server .exe — relative path from install folder
Default Port	Yes	Game port
Default Query Port	Yes	Steam Query port
Default Steam Port	No	Steam network port (0 = not used)
Max Players	Yes	Default player count
Start Arguments	Yes	Command-line args (supports placeholder variables)
Stop Command	No	Graceful shutdown command (blank = kill process)

Placeholder variables for Start Arguments

{ip} → ServerIp
{port} → ServerPort
{qport} → QueryPort
{name} → ServerName
{max} → MaxPlayers
{map} → GameSpecificSettings["map"]
{password} → ServerPassword

24.2 Import / Export plugins

Plugins can be exported to a .cs source file and imported on another machine.

- **Export:** Tools → Export Plugin → select a plugin → save as .cs file
- **Import:** Tools → Import Plugin → select a .cs file — the plugin is compiled in memory and registered immediately without restarting WGS
- Exported files are self-contained C# source files — inspect, edit and share freely

25. Troubleshooting

? Server won't start — 'executable not found'	Game files are not installed or the executable path is wrong. Re-run 'Install / Update', or check manually that the .exe exists in the install folder.
? SteamCMD fails — ERROR! App state is 0x...	Usually a network issue or Steam servers are temporarily down. Try again shortly. If login is required, verify your Steam Guard code.
? Port already in use (pre-flight warning)	Another program is using the same port. Change the port in Settings or stop the conflicting program.
? Wreckfest crashes on startup	Wreckfest needs UseNativeConsole=true — already set in the plugin. Verify the install path and that server_config.cfg exists.
? RCON won't connect	Confirm the RCON port and password match the server's config file and that a firewall rule exists for that port.
? Backup fails	Disk space may be full or WGS lacks write permission to %APPDATA%. Check the Windows Event Log for details.
? Discord bot won't connect	Verify the bot token and channel ID are correct. The bot needs Read Messages and Send Messages permissions in that channel. Check the bot status in Settings.
? Web API returns 401	The request is missing the token. Add Authorization: Bearer header or append ?token= to the URL.
? Auto-restart disabled — crash limit reached	The server crashed too many times in 10 minutes. Fix the underlying problem, then re-enable Auto Restart in the server's Settings tab.
? App won't launch — .NET error	Install .NET 8 Runtime: dotnet.microsoft.com/download/dotnet/8.0
? Server processes survive after WGS closes	Always stop servers with the Stop button before closing WGS. Force-closing WGS via Task Manager may leave server processes running.

26. Keyboard Shortcuts

Shortcut	Action
Ctrl + V	Paste (all text fields)
Ctrl + C	Copy (all text fields)
Ctrl + X	Cut (all text fields)
Ctrl + A	Select all (text fields)
Enter	Send console / RCON command

27. File Locations

WGS stores configuration in Windows AppData and writes nothing permanent to its own installation directory.

Path	Contents
%APPDATA%\WGS\servers.json	All server configurations
%APPDATA%\WGS\settings.json	Global application settings
%APPDATA%\WGS\notifications.json	Discord notification settings
%APPDATA%\WGS\scheduled_tasks.json	Scheduled task definitions
%APPDATA%\WGS\custom_plugins.json	Custom Plugin Creator entries
%APPDATA%\WGS\player_stats.db	Player session history (SQLite)
%APPDATA%\WGS\steamcmd\	SteamCMD installation (auto-downloaded)
%APPDATA%\WGS\backups\	Server backup zip files

Config backup tip

Copy %APPDATA%\WGS\servers.json before reinstalling WGS — all server definitions are stored in that one file.